For example, under */usr/local/etc*, you will see that the *keepalived*, *rc.d*, and *sysconfig* directories have been created. Copy the files as follows:

```
root@Firewall1 etc]# cp rc.d/init.d/keepalived /etc/rc.d/init.d/
[root@Firewall1 etc]# cp sysconfig/keepalived /etc/sysconfig/
```

Note: Do not currently copy the original keepalived.conf file that was generated at the time of the keepalived package installation. We will use a different keepalived configuration, which will be produced by the conntrack daemon.

For better administration, if you wish, add the *keepalived* daemon in the *chkconfig* list:

```
[root@Firewall1 etc]# chkconfig --add keepalived
```

conntrack: The third and last item to install is the *conntrackd* daemon. *Conntrack* is a utility that maintains the packet state, and allows systems administrators to interact (from user-space) with the in-kernel connection-tracking system. *Conntrack* works with *iptables*, which then maintains the packet state between cluster members, and any modification in these is notified to others using the UDP protocol.

The *conntrack-tools* package contains two programs:

- *conntrack* is a command-line utility that gives you complete control over the connection-tracking system. With *conntrack*, you can modify and query the existing state entries; you can also observe the real-time entries.
- *conntrackd* is the connection-tracking daemon. It is used to deploy highly-available firewalls, but you can also use it to collect the statistics of the firewall connection table use, which can be analysed later. (http://conntrack-tools.netfilter.org/manual.html)

In order to get *conntrack-tools* working, you must have Linux kernel version >= 2.6.18; one that at least has support for the following modules (I have shown how to validate the modules in the 'Verifying kernel support' section):

- The connection tracking system:
  - CONFIG_NF_CONNTRACK……….checks if the said module is present.
  - CONFIG_NF_CONNTRACK_IPV4…checks if the said module is present
  - CONFIG_NF_CONNTRACK_IPV6….confirms if the said module is present (if your set-up supports IPv6)
- *nfnetlink*: The common messaging interface for Netfilter
  - CONFIG_NETFILTER_NETLINK= module
  - CONFIG_NF_CT_NETLINK= module
  - CONFIG_NF_CONNTRACK_EVENTS=y

Download the *conntrackd* tar file from http://conntrack-tools.netfilter.org/downloads.html.

Conntrack-tools requires *libnfnetlink* and *libnetfilter_conntrack*; you can get them from netfilter.org.

Before proceeding further, it is imperative to understand the role that *libnfnetlink* and *libnetfilter_conntrack* play for the *conntrackd* daemon (and to compile and install them).

libnfnetlink

libnfnetlink is the low-level library for *netfilter*. It facilitates a common messaging infrastructure for in-kernel netfilter subsystems. The prerequisite for *libnfnetlink* is a kernel that features the *nfnetlink* subsystem. All the 2.6.x kernels with version >= 2.6.14 natively support the above modules. Untar the downloaded *libnfnetlink*:

```
[root@Firewall1 hacluster]# tar -jxvf libnfnetlink-1.0.0.tar.bz2
```

Compile and install the package:

```
[root@Firewall1 libnfnetlink-1.0.0]# ./configure && make && make install
```

If this goes smoothly, it's now time to proceed to *libnetfilter_conntrack*.

libnetfilter_conntrack

This is a user-space library that provides an interface to the in-kernel connection-tracking system. The prerequisite for *libnetfilter_conntrack* is that the Linux kernel version should be >= 2.6.18 and the kernel should have support enabled for:

- The connection tracking system
- nfnetlink
- ctnetlink
- The connection tracking event

Note: Fedora users: By default, the installation adds the libraries in */usr/local/*, which most Fedora systems are not aware of. Hence, you might run into issues while compiling the next package. So, before you proceed further, make sure you perform the following steps, to enable the system to detect the shared libraries from the said path:

```
root@Firewall1 libnetfilter_conntrack-0.0.101]# PKG_CONFIG_PATH=/usr/local/
lib/pkgconfig;export PKG_CONFIG_PATH
```

Next, edit the */etc/ld.so.conf* file, add */usr/local/lib*, save the file and run the *ldconfig* command:

```
root@Firewall1 libnetfilter_conntrack-0.0.101]# ldconfig
```